Key Mathematical Theorems and Their Application to k-Uniform Tessellations

Theorem / Lemma	Statement	Implication for k-uniform Tessellations	Reference
Floquet-Bloch	Eigenfunktionen haben Form $\psi_k(r) = e^{ik \cdot r} u_i$	IDS can be computed via Brillouin zone integration	Floquet (1883)
Weyl Law	$N(E) \sim (\mathrm{Vol}/4\pi^2)E$ as $E \to \infty$	Asymptotic density of states is universal	Weyl (1911)
Periodicity	Gittervektoren a_1, a_2 definieren Symmetri	Wallpaper group structure is preserved	Bravais (1850)
Spectral Gap	Bandgaps existieren wenn Gitterperiode e	k-uniform gitter können Bandgaps haben	Kittel (1996)
Euler Char.	$V - E + F = 2$ für geschlossene Flächen	Topological constraints auf FD-Struktur	Euler (1758)
Orbit-Stabilizer	$|Aut(G)| = |Orbit| × |Stabilizer|	k Orbits führen zu k verschiedenen Vertex-Typen	Burnside (1897)
Symmetry Groups	Genau 17 Wallpaper-Gruppen in $\mathbb{R}^2$	k-uniform tessellationen nutzen diese 17 Gruppen	Fedorov (1891)